

Q1. Smart Home Automation System – Intelligent Agents

Introduction

An intelligent agent is a system that observes the environment through sensors and performs actions using actuators to achieve specific goals. In a smart home, intelligent agents control lighting, temperature, and security by responding to user behavior and environmental conditions.

1. Simple Reflex Agent

Definition:

A Simple Reflex Agent acts only on the current input using predefined IF–THEN rules. It does not store previous information.

Example:

- If motion is detected → Turn ON the lights.
- If smoke is detected → Activate the alarm.

Advantages:

- Fast response.
- Easy to implement.

Disadvantages:

- Cannot learn user habits.
- Cannot make decisions based on past events.

2. Model-Based Agent

Definition:

A Model-Based Agent keeps an internal model of the environment and uses both current and previous information for decision-making.

Example:

- Turns OFF lights if no motion is detected for 20 minutes.
- Maintains room temperature based on previous readings.

Advantages:

- Works in changing environments.
- Makes better decisions than a reflex agent.

Disadvantages:

- Requires more memory and processing.

3. Goal-Based Agent

Definition:

A Goal-Based Agent selects actions that help achieve a specific goal.

Example:

- Maintains room temperature at **24°C** while reducing electricity consumption.

Advantages:

- Flexible.
- Can choose the best action to reach a goal.

Disadvantages:

- Takes more time to evaluate possible actions.

4. Utility-Based Agent

Definition:

A Utility-Based Agent selects the action that provides the highest overall benefit by considering multiple factors.

Example:

- Balances user comfort, energy efficiency, and security.
- Lowers AC usage when electricity cost is high.

Advantages:

- Produces optimal decisions.
- Considers multiple objectives.

Disadvantages:

- Utility calculation is complex.

Comparison

Agent Type	Memory	Goal-Oriented	Learns	Example
Simple Reflex	No	No	No	Turn ON light when motion detected
Model-Based	Yes	No	Partial	Turn OFF light after inactivity
Goal-Based	Yes	Yes	No	Maintain desired room temperature
Utility-Based	Yes	Yes	Yes	Balance comfort, energy, and security

Best Approach

A **Hybrid Agent** combining **Model-Based**, **Goal-Based**, and **Utility-Based** approaches is the most suitable because it:

- Learns user preferences.
- Saves energy.
- Maintains comfort.
- Improves home security.
- Makes intelligent real-time decisions.

Conclusion

A hybrid intelligent agent provides the best performance in a smart home by ensuring comfort, energy efficiency, and security through intelligent decision-making.

Q2. Robot Maze – Uninformed Search Strategies

Introduction

A robot placed in an unknown maze must find the exit without prior knowledge of the layout. Uninformed search algorithms explore the maze without using heuristic information. Two common algorithms are **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**.

1. Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the lowest cumulative path cost first.

Working

- Start from the initial position.
- Expand the least-cost path.
- Continue until the exit is found.

Advantages

- Finds the optimal path.
- Works well when path costs are different.

Disadvantages

- Requires high memory.
- Slower for large search spaces.

Complexity

- **Time Complexity:** Exponential
- **Space Complexity:** Exponential

2. Iterative Deepening Search (IDS)

Definition

IDS repeatedly performs Depth First Search with increasing depth limits until the goal is found.

Working

- Search depth 0.
- If goal is not found, increase depth to 1.
- Repeat until the exit is reached.

Advantages

- Low memory usage.
- Finds the shortest path when all costs are equal.

Disadvantages

- Repeats node expansion.
- Can be slower because searches are repeated.

Complexity

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(bd)$

Comparison

Feature	UCS	IDS
Uses Path Cost	Yes	No
Optimal Solution	Yes	Yes (Equal Costs)
Memory Usage	High	Low
Speed	Slower	Faster in low-memory systems

Relation to Intelligent Agents

Simple Reflex Agent

- Reacts only to the current state.
- Not suitable for maze solving.

Model-Based Agent

- Remembers visited paths.
- Avoids revisiting the same locations.

Goal-Based Agent

- Plans actions to reach the maze exit.

Utility-Based Agent

- Selects the most efficient and least costly path.

Best Combination

Goal-Based Agent + Uniform Cost Search (UCS)

Justification

- Goal-Based Agent focuses on reaching the exit.
- UCS guarantees the least-cost path.
- Suitable for unknown mazes with different path costs.
- Produces accurate and efficient navigation.

Conclusion

For solving an unknown maze, the combination of a **Goal-Based Agent** and **Uniform Cost Search (UCS)** is the most effective because it provides optimal path selection, efficient decision-making, and reliable navigation.